Last iterate convergence and entropy limits of optimistic matrix multiplicative weights

Abstract

We prove last iterate convergence of raw optimistic matrix multiplicative weights in finite-dimensional bilinear quantum zero-sum games in two regimes. In arbitrary complex matrix dimensions, existence of one strictly complementary Nash pair suffices for both entire state sequences to converge to one Nash pair with the same supports. Forward relative entropy from that selected limit tends to zero; the strict witness need not be the selected pair. For one qubit per player, state convergence holds without strict complementarity, uniqueness or a full-rank equilibrium. Both results use maximally mixed initialization and one sufficiently small fixed step. The arbitrary-dimensional proof combines an optimistic entropy estimate with a unitary commutator bound and controlled spectral leakage at singular limits. On a fixed non-strict qubit game from the literature, all four forward relative entropies instead converge to the same strictly positive limit despite state convergence. An exact score ratio gives an explicit positive lower bound. On that same orbit, state error has exact order $k^{-1/3}$ and Nash gap has exact order k^{-1}, with leading constants determined by the positive entropy limit. A fixed qutrit–qubit example further selects a mixed state on a nonunique singular Nash face despite the absence of any strictly complementary pair; both selected forward entropies vanish. No rate or selection formula uniform over games is asserted.

Keywords: quantum zero-sum games; optimistic matrix multiplicative weights; last iterate convergence; singular equilibria; strict complementarity

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1 The exact algorithm and convergence statement

Let $\mathcal{H}_2$ be the real Hilbert space of complex Hermitian two-by-two matrices, with pairing $\langle X, Y\rangle = \mathrm{Tr}(XY)$, and set

$$\mathcal{D}_2 = \{X \in \mathcal{H}_2 : X \succeq 0,\ \mathrm{Tr}\, X = 1\}, \qquad \mathcal{Z} = \mathcal{D}_2 \times \mathcal{D}_2. \tag{1}$$

Fix a Hermitian four-by-four observable U with $\|U\|_{\mathrm{op}} \le 1$. Alice minimizes and Bob maximizes

$$f(X, Y) = \mathrm{Tr}\big(U(X \otimes Y)\big). \tag{2}$$

Write $F = (\nabla_X f, -\nabla_Y f)$, with full Hermitian-space Frobenius gradients. Choose any $L \ge 1$ satisfying

$$\|F(z) - F(z')\|_{F,\mathrm{prod}} \le L\|z - z'\|_{F,\mathrm{prod}} \qquad (z, z' \in \mathcal{Z}), \qquad 0 < \eta < \frac{1}{10L}. \tag{3}$$

The step η is fixed. For a positive definite density matrix Q, put

$$P_\eta(Q, G) = \frac{\exp(\log Q - \eta G)}{\operatorname{Tr} \exp(\log Q - \eta G)}. \tag{4}$$

For a pair, apply this formula separately to the two blocks, with separate trace normalizers. The iteration is

$$\begin{aligned} z_{k+1} &= P_\eta(\widehat{z}_k, F(z_k)), \\ \widehat{z}_{k+1} &= P_\eta(\widehat{z}_k, F(z_{k+1})), \\ z_0 &= \widehat{z}_0 = (I_2/2, I_2/2). \end{aligned} \tag{5}$$

It uses one new evaluation of F per round after the initial evaluation. The entropy defines the prox geometry; no entropy penalty is added to the payoff.

Theorem 1. *Every finite state in* (5) *is positive definite and trace one. There is a single Nash pair $z_\infty = (X_\infty, Y_\infty) \in \mathcal{Z}$ such that*

$$z_k \longrightarrow z_\infty, \qquad \widehat{z}_k \longrightarrow z_\infty \quad \textit{in the product Frobenius norm.} \tag{6}$$

In particular

$$f(X_\infty, Y) \le f(X_\infty, Y_\infty) \le f(X, Y_\infty) \qquad (X, Y \in \mathcal{D}_2). \tag{7}$$

The equilibrium set may be nonunique, its states may be singular, and its complementarity slacks may vanish. No invariant commuting basis or initialization near an equilibrium is assumed.

The theorem concerns the stated raw recurrence and maximally mixed played and auxiliary initialization. It gives no general convergence rate, finite total length or prescribed entropy-selected equilibrium. Forward relative entropy to the limiting pair need not tend to zero, as Proposition 6.1 shows. Theorem 1 is specific to qubits. Section 7 proves a separate arbitrary-dimensional conclusion under existence of one strictly complementary equilibrium. Section 8 gives a separate fixed qutrit–qubit example with a nonunique Nash face and no strictly complementary pair. Neither theorem covers arbitrary two-point initialization or nonlinear payoffs.

1.1 Relation to existing results

The algorithm is the entropy instance of the optimistic quantum methods studied by Vasconcelos et al. [1]. Their result gives an averaged equilibrium-gap guarantee. Barakat et al. [2, Theorem 11] give related averaged guarantees in a broader symmetric-cone and nonlinear-game setting. Averaged convergence does not imply (6).

For products of probability simplexes, Orabona [3, Theorem 2] proves last iterate convergence without uniqueness or strict complementarity. We use the generic optimistic

entropy estimate underlying that argument, but not its simplex-specific boundary proof. In the matrix setting, convergence in norm to a singular state does not by itself imply convergence of the forward relative entropy. The spectral reciprocity issue is already documented by López and Papa Quiroz [4, Remark 4]. Weis [8, Example 2.2] gives a qubit boundary discontinuity example, and Weis and Knauf [9, Example 11] give full-rank paths approaching a pure state with a positive limiting forward entropy. These are geometric paths, rather than the initialized optimistic orbit studied in Section 6.

A rate claim would be substantially different. Su, Vlatakis-Gkaragkounis and Xiong [5, Theorem 5.2] construct a fixed qubit game with a unique strictly complementary equilibrium on which raw OMMWU has only polynomial decay from the maximally mixed start. Their positive convergence examples are consistent with the qualitative conclusion here. Entropy-regularized QRE arguments, such as the displayed update analyzed in [6, Appendix C.5], concern a different recurrence and target. The proofs below retain the unregularized update throughout. The arbitrary-dimensional result in Section 7 uses only existence of one strictly complementary pair; it does not require the every-equilibrium strict-complementarity hypothesis used in the geometric separation theorem of [6, Theorem 9]. It also does not invoke that source's regularized recurrence or a local warm start.

2 Bloch coordinates and the entropy estimate

Let $\sigma = (\sigma_x, \sigma_y, \sigma_z)$ be the three Pauli matrices. Every matrix in $\mathcal{D}_2$ is uniquely

$$\rho(x) = \frac{I_2 + x \cdot \sigma}{2}, \qquad x \in B := \{x \in \mathbb{R}^3 : |x| \le 1\}, \qquad \|\rho(x) - \rho(x')\|_F^2 = \frac{1}{2}|x - x'|^2. \tag{8}$$

All three coordinates are retained, including the imaginary off-diagonal one. The payoff becomes

$$f(x, y) = c + a \cdot x + b \cdot y + x \cdot Ay, \qquad g(x, y) = (a + Ay, -b - A^T x), \tag{9}$$

where A is real three-by-three. Scalar identity components of the matrix gradients cancel in the separate normalizers, and their traceless coefficients are exactly g. Put $\ell = \|A\|_{\mathrm{op}}$. Retaining just the traceless parts in (3) gives $2\ell \le L$.

For $x \in B$, define

$$h(x) = \frac{1 + |x|}{2} \log \frac{1 + |x|}{2} + \frac{1 - |x|}{2} \log \frac{1 - |x|}{2}, \qquad 0 \log 0 = 0. \tag{10}$$

In the open ball its gradient and inverse are

$$\theta(x) = \operatorname{atanh}(|x|) \frac{x}{|x|}, \qquad T(\theta) = \frac{\tanh |\theta|}{|\theta|} \theta, \qquad h^*(\theta) = \log(2 \cosh |\theta|). \tag{11}$$

Use continuous values at zero. The radial Hessian eigenvalue of h is $(1 - |x|^2)^{-1}$ and the tangential ones are $\operatorname{atanh}(|x|)/|x|$. Hence h is 1-strongly convex. Its Bregman divergence

D is one-player quantum relative entropy; it extends to a boundary comparator and satisfies

$$D(u,v) \geq \frac{1}{2}|u-v|^2 \qquad (u \in B,\ |v|<1). \tag{12}$$

For a pair, D denotes the sum of the two divergences.

Write $\alpha_k = \theta(x_k)$, $\beta_k = \theta(y_k)$ for played duals, and use hats for auxiliary duals. Equation (5) becomes exactly

$$\begin{aligned}(\alpha_{k+1},\beta_{k+1}) &= (\widehat{\alpha}_k,\widehat{\beta}_k) - \eta g(x_k,y_k),\\ (\widehat{\alpha}_{k+1},\widehat{\beta}_{k+1}) &= (\widehat{\alpha}_k,\widehat{\beta}_k) - \eta g(x_{k+1},y_{k+1}),\end{aligned} \tag{13}$$

with all initial duals zero. Finite duals give positive definite states. Since g is bounded on $B \times B$, all played and auxiliary dual norms are $O(k+1)$. For

$$t(r) = \frac{\tanh r}{r}, \quad t(0)=1, \tag{14}$$

the inequality $t(r) \geq (1+r)^{-1}$ then gives a fixed $\kappa > 0$ with

$$t(|\alpha_k|),\ t(|\beta_k|) \geq \frac{\kappa}{k+1}. \tag{15}$$

These bounds concern the actual initialized trajectory.

2.1 A comparator energy

For this subsection write $z_k = (x_k, y_k)$ and $v_k = \widehat{z}_k$ in Bloch coordinates. For a Nash pair p_*, define

$$s = \eta\ell, \quad \mathcal{A}_k = D(v_k, z_k), \quad \mathcal{B}_{k+1} = D(z_{k+1}, v_k), \quad E_k(p_*) = D(p_*, v_k) + s\mathcal{A}_k. \tag{16}$$

Here $s < 1/20$; if $A = 0$, use $s = 0$.

Lemma 1. *For every Nash pair p_* and $k \geq 0$,*

$$\begin{aligned}E_{k+1}(p_*) \leq E_k(p_*) &- (1-3s)\mathcal{A}_{k+1} - (1-s)\mathcal{B}_{k+1}\\ &- \eta\langle g(z_{k+1}), z_{k+1} - p_*\rangle.\end{aligned} \tag{17}$$

Consequently $D(p_*, v_k)$ has a finite limit, the two Bregman-step sums are finite, and

$$|z_k - v_k| \to 0, \qquad |z_{k+1} - v_k| \to 0, \qquad |z_{k+1} - z_k| \to 0. \tag{18}$$

Moreover

$$\sum_{k\geq 1} \langle g(z_k), z_k - p_*\rangle < \infty. \tag{19}$$

Proof. The two exact dual updates and the three-point identity give

$$D(p_*, v_{k+1}) = D(p_*, v_k) - \mathcal{A}_{k+1} - \mathcal{B}_{k+1} + \eta\langle g(z_k) - g(z_{k+1}), v_{k+1} - z_{k+1}\rangle - \eta\langle g(z_{k+1}), z_{k+1} - p_*\rangle. \tag{20}$$

Let $a = |v_{k+1} - z_{k+1}|$, $b = |z_{k+1} - v_k|$ and $c = |v_k - z_k|$. The cross term is at most $sa(b + c)$, hence at most $s(a^2 + b^2/2 + c^2/2) \le s(2\mathcal{A}_{k+1} + \mathcal{B}_{k+1} + \mathcal{A}_k)$. This proves (17). The final pairing is nonnegative for a Nash comparator. Summation and (12) give the stated summability and regularity. Since $\mathcal{A}_k \to 0$, convergence of $E_k(p_*)$ gives convergence of $D(p_*, v_k)$. ∎

This is the standard optimistic mirror estimate, with a finite entropy comparator; no global Bregman diameter or boundary reciprocity is used. One further exact identity will matter:

$$(\widehat{\alpha}_k, \widehat{\beta}_k) - (\alpha_k, \beta_k) = -\eta\big(g(z_k) - g(z_{k-1})\big) \longrightarrow 0 \qquad (k \ge 1). \tag{21}$$

3 Entropy bounds and equilibrium ranks

The rest of the proof uses Bloch coordinates.

Lemma 2. *Let $p \in B$ be pure, so $|p| = 1$. Then*

$$D(p, T(\theta)) = \log(2\cosh|\theta|) - p \cdot \theta. \tag{22}$$

Boundedness of this divergence and $|\theta_k| \to \infty$ imply $T(\theta_k) \to p$. If $|q| < 1$, bounded $D(q, T(\theta_k))$ instead implies bounded θ_k. There is a numerical constant C_0 such that

$$D(p, x) \le C_0(1 + |\theta(x)|)(1 - p \cdot x) \qquad (|x| < 1). \tag{23}$$

For any fixed comparator in B, its divergence as a function of the dual second argument is 2-Lipschitz.

Proof. The conjugate formula gives (22). Since $\log(2\cosh r) \ge r$, bounded divergence implies bounded $|\theta_k| - p \cdot \theta_k$. When the norms diverge, their directions tend to p. For an interior comparator,

$$D(q, T(\theta)) \ge h(q) + (1 - |q|)|\theta|,$$

which proves the dual bound.

To prove (23), put $r = |\theta|$ and $n = \theta/r$. The relevant identities are

$$\begin{aligned} D(p, T(\theta)) &= \log(1 + e^{-2r}) + r(1 - p \cdot n), \\ 1 - p \cdot T(\theta) &= 1 - \tanh r + \tanh r(1 - p \cdot n). \end{aligned} \tag{24}$$

For $r \ge 1$, use $\log(1 + e^{-2r}) \le 1 - \tanh r$ and $\tanh r \ge \tanh 1 > 1/2$. For $r \le 1$, the second line is at least $1 - \tanh 1 > 0$, while the first is bounded. For example $C_0 = 16$

suffices. Finally $D(q, T(\theta)) = h(q) + h^*(\theta) - q \cdot \theta$ has gradient $T(\theta) - q$, of norm at most two. ∎

The Nash set is a Cartesian product $\mathcal{X}^* \times \mathcal{Y}^*$ of nonempty convex compact optimal-strategy sets. Strict convexity of the Euclidean ball implies that each factor either contains an interior point or is a singleton consisting of a pure point. Choose a Nash pair (p, q) with an interior component whenever its marginal optimal set admits one. A chosen boundary component is therefore the unique optimum of its player. The first-order conditions are

$$a + Aq = -\lambda p, \qquad b + A^T p = \mu q, \tag{25}$$

where $\lambda = 0$ if p is interior, $\mu = 0$ if q is interior, and otherwise the corresponding coefficients are nonnegative. The saddle pairing in (19) is exactly

$$\langle g(x_k, y_k), (x_k - p, y_k - q) \rangle = \lambda(1 - p \cdot x_k) + \mu(1 - q \cdot y_k), \tag{26}$$

with zero-coefficient terms omitted.

If both marginal sets contain interior points, Lemmas 1 and 2 bound all duals and keep the states uniformly inside the ball. The primal regularity (18) and continuity of ∇h then give $g(z_k) \to 0$ from (13). Every cluster point z_* is Nash. Along a subsequence tending to it, $D(z_*, v_k) \to 0$ by ordinary interior continuity. This divergence has a limit by Lemma 1, so its full limit is zero, and (12) proves convergence of both sequences. It remains to treat the boundary configurations.

4 One pure optimal block

Suppose $\mathcal{X}^* = \{p\}$ with $|p| = 1$, while $\mathcal{Y}^*$ contains an interior point q. Here $\mu = 0$ in (25). We first show that q can be chosen interior with $\lambda > 0$; this is a consequence of the geometry, not an extra assumption.

If its initial slack is zero, both $a + Aq = 0$ and $b + A^T p = 0$. If $p \notin \operatorname{range} A$, its orthogonal projection onto $\operatorname{range} A$ is interior and has the same A^T-image. It would be another optimal Alice strategy, a contradiction. Thus choose w with $Aw = p$. For sufficiently small $\varepsilon > 0$, the point $q - \varepsilon w$ is still interior and gives $a + A(q - \varepsilon w) = -\varepsilon p$. Bob's gradient at p remains zero. Replacing q by this Nash strategy makes $\lambda > 0$.

Fix this choice. By (19) and (26),

$$\delta_i = 1 - p \cdot x_i, \qquad D_k = \sum_{i=1}^{k} \delta_i \longrightarrow D_\infty < \infty. \tag{27}$$

It follows that $x_k, \widehat{x}_k \to p$. The Bob duals stay bounded because the comparator q is interior.

4.1 An exact bilinear functional

Put $P = I - pp^T$ and $C = PA$. The cumulative identities from (13) are

$$\begin{aligned} u_k &= P\widehat{\alpha}_k = -\eta C \sum_{i=1}^{k} (y_i - q), \\ r_k &= \widehat{\beta}_k + \eta A^T p D_k = \eta C^T \sum_{i=1}^{k} P x_i. \end{aligned} \tag{28}$$

Thus $u_k \in \operatorname{range} C$ and $r_k \in \operatorname{range} C^T$. The sequence r_k is bounded. Also $u_k = o(k)$: the auxiliary Alice state tends to p, its dual direction tends to p, and its dual norm is $O(k+1)$.

Let C^+ denote the Moore–Penrose inverse, including when $C = 0$, and define

$$S_k = (C^+ u_k) \cdot r_k. \tag{29}$$

Then $S_k = o(k)$. Expanding the increment using the new u_i and old r_{i-1} gives the exact identity

$$S_i - S_{i-1} = -\eta (y_i - q) \cdot r_{i-1} + \eta u_i \cdot P x_i. \tag{30}$$

Indeed C^+C projects onto $\operatorname{range} C^T$, while CC^+ fixes u_i.

The Cesàro average of the final term tends to zero. By (21), $u_i = P\alpha_i + o(1)$, and

$$(P\alpha_i) \cdot P x_i = \frac{\operatorname{atanh} |x_i|}{|x_i|} |P x_i|^2 \le 2(1 + |\alpha_i|)\delta_i \le K(i+1)\delta_i. \tag{31}$$

We used $r / \tanh r \le 1 + r$ and $|P x_i|^2 \le 2\delta_i$. Summability in (27) implies

$$\frac{1}{n} \sum_{i=1}^{n} (i+1)\delta_i \longrightarrow 0. \tag{32}$$

For a direct verification, split at a fixed N: the finite head vanishes after division by n, and the remaining part is bounded by a constant times $\sum_{i>N} \delta_i$. Summing (30) therefore yields

$$\frac{1}{n} \sum_{i=1}^{n} (y_i - q) \cdot r_{i-1} \longrightarrow 0. \tag{33}$$

4.2 The limiting Nash comparator

Set

$$e = -\eta A^T p D_\infty, \qquad q^\dagger = \operatorname*{arg\,min}_{\substack{y \in B \\ Cy = Cq}} \{h(y) - e \cdot y\}. \tag{34}$$

This minimizer exists and is unique by compactness and strict convexity. It is interior: the slice contains the interior point q, and at any boundary point the directional derivative of h toward q is $-\infty$, whereas the linear term has finite derivative. Hence

$$\nabla h(q^\dagger) - e \in \operatorname{range} C^T. \tag{35}$$

The played Bob update, $x_{i-1} \to p$ and $D_{i-1} \to D_\infty$ give

$$r_{i-1} - (\nabla h(y_i) - e) \longrightarrow 0. \tag{36}$$

Because $C(q^\dagger - q) = 0$, equation (33) also holds with $q^\dagger$ in place of q. Moreover (28) and $u_n = o(n)$ imply

$$C\left(\frac{1}{n}\sum_{i=1}^{n} y_i - q^\dagger\right) \longrightarrow 0. \tag{37}$$

Subtracting the average pairing with (35) and using (36), we obtain

$$\frac{1}{n}\sum_{i=1}^{n}\langle \nabla h(y_i) - \nabla h(q^\dagger), y_i - q^\dagger\rangle \longrightarrow 0.$$

Strong convexity gives

$$\frac{1}{n}\sum_{i=1}^{n} |y_i - q^\dagger|^2 \longrightarrow 0, \qquad \frac{1}{n}\sum_{i=1}^{n} y_i \longrightarrow q^\dagger. \tag{38}$$

The point $(p, q^\dagger)$ must now be checked to be Nash. The exact accumulated Alice score satisfies

$$\frac{\widehat{\alpha}_n}{n} = \eta\lambda p - \eta A\left(\frac{1}{n}\sum_{i=1}^{n} y_i - q\right) \longrightarrow \eta\big[\lambda p - A(q^\dagger - q)\big]. \tag{39}$$

The limit is a multiple of p by the affine constraint in (34). Its coefficient is nonnegative, since a negative one would force $\widehat{x}_n \to -p$, contradicting $\widehat{x}_n \to p$. Thus $a + Aq^\dagger = -\lambda^\dagger p$ for some $\lambda^\dagger \geq 0$. Together with $b + A^T p = 0$, this proves the saddle conditions.

Finally (23), the linear dual bound and (32) show that the Cesàro mean of $D(p, x_i)$ tends to zero. The same holds for $D(p, \widehat{x}_i)$ by (21) and the dual Lipschitz bound in Lemma 2. The Bob states stay in a compact subset of the open ball, so (38), continuity and (18) give vanishing Cesàro mean of $D(q^\dagger, \widehat{y}_i)$.

The sum $D((p, q^\dagger), \widehat{z}_i)$ has an ordinary limit by Lemma 1, since its comparator has now been proved Nash. Its Cesàro mean is zero, so the ordinary limit is zero. Strong convexity proves convergence of both entire sequences to $(p, q^\dagger)$. The opposite mixed-rank configuration follows by exchanging players and changing the payoff sign. The point in (34) depends on D_∞; it is not a theorem-wide initial-entropy selection rule.

5 Two pure optimal blocks

Suppose now $\mathcal{X}^* = \{p\}$ and $\mathcal{Y}^* = \{q\}$, with $|p| = |q| = 1$. If both slacks in (25) are positive, (19) immediately gives $x_k \to p$ and $y_k \to q$. We handle the vanishing-slack cases through exact score identities.

5.1 One positive slack

Assume $\lambda > 0$ and $\mu = 0$. We already know

$$\sum_{i\geq 1}(1 - p \cdot x_i) < \infty, \qquad x_i \to p. \tag{40}$$

Put $P = I - pp^T$ and $C = PA$. The slice

$$\{y \in B : Cy = Cq\} \tag{41}$$

is the singleton $\{q\}$. Indeed, a short segment from q toward a different point of this slice lies in the open ball, keeps $a + Ay$ parallel to p, and retains its strictly negative radial coefficient. Since $b + A^T p = 0$, each such point would be another optimal Bob strategy.

It follows that $q \in \operatorname{range} C^T$: otherwise its orthogonal projection onto $\operatorname{range} C^T$ would have norm less than one and the same C-image. Define

$$d = (C^T C)^+ q, \qquad w = Cd, \qquad \gamma = p \cdot Ad. \tag{42}$$

Then $C^T w = q$, $w \perp p$ and $Ad = w + \gamma p$. Set

$$\varepsilon_i = 1 - q \cdot y_i, \qquad H_k = \sum_{j=1}^{k} \varepsilon_j, \qquad \varepsilon_0 = 1. \tag{43}$$

Since $w \cdot (a + Ay) = q \cdot y - 1$, the actual played score satisfies

$$w \cdot \alpha_i = \eta(H_{i-1} + \varepsilon_{i-1}), \qquad i \geq 1. \tag{44}$$

The bracket is uniformly positive: at $i = 1$ it equals one, and for $i \geq 2$ it includes $\varepsilon_1 > 0$ because the first played state is interior. This is where the canonical zero dual initialization is used.

The auxiliary Bob increment now obeys

$$\begin{aligned} d \cdot (\widehat{\beta}_i - \widehat{\beta}_{i-1}) &= \eta(Ad) \cdot (x_i - p) \\ &= \eta^2 t(|\alpha_i|)(H_{i-1} + \varepsilon_{i-1}) - \eta\gamma(1 - p \cdot x_i). \end{aligned} \tag{45}$$

The first term has a divergent harmonic sum by (15); the second has an absolutely convergent sum by (40). Thus $d \cdot \widehat{\beta}_k \to +\infty$ and $|\widehat{\beta}_k| \to \infty$. Bounded $D(q, \widehat{y}_k)$ and Lemma 2 give $\widehat{y}_k \to q$. Equation (18) gives the same played limit. The case $\lambda = 0$, $\mu > 0$ follows by symmetry.

5.2 Both slacks zero

Now

$$a = -Aq, \qquad b = -A^T p. \tag{46}$$

We first establish the needed range geometry. One has $p \in \operatorname{range} A$ and $q \in \operatorname{range} A^T$. For instance, projecting p onto $\operatorname{range} A$ preserves its A^T-image; if the projection differed from p, it would give an interior Alice strategy with zero signed gradients against q, hence another Nash pair.

The Moore–Penrose inverse is the inverse from range A to range A^T on these subspaces. Put

$$\tau = q \cdot A^+ p = p \cdot (A^T)^+ q. \tag{47}$$

Necessarily $\tau = 0$. If $\tau > 0$, then $q - \varepsilon A^+ p$ is interior for small $\varepsilon > 0$ and gives Alice gradient $-\varepsilon p$, producing another Nash Bob strategy. If $\tau < 0$, then $p + \varepsilon (A^T)^+ q$ is interior and gives Bob gradient εq, producing another Nash Alice strategy. Both contradict the singleton marginal sets.

Define

$$\begin{aligned} w &= A^+ p, \qquad v = (A^T)^+ q, \\ d_x &= (AA^T)^+ p, \qquad d_y = (A^T A)^+ q. \end{aligned} \tag{48}$$

The identities needed below are

$$Aw = p, \quad A^T v = q, \quad A^T d_x = w, \quad A d_y = v, \quad q \cdot w = p \cdot v = 0. \tag{49}$$

They hold also for rank-deficient A, by its singular-value decomposition and the range memberships just proved.

Let

$$\begin{aligned} \delta_i &= 1 - p \cdot x_i, \qquad \varepsilon_i = 1 - q \cdot y_i, \qquad \delta_0 = \varepsilon_0 = 1, \\ D_k &= \sum_{j=1}^{k} \delta_j, \qquad H_k = \sum_{j=1}^{k} \varepsilon_j. \end{aligned} \tag{50}$$

Exact summation of (13), including the extra played step, gives

$$w \cdot \beta_i = -\eta (D_{i-1} + \delta_{i-1}), \qquad v \cdot \alpha_i = \eta (H_{i-1} + \varepsilon_{i-1}). \tag{51}$$

Both bracketed quantities are uniformly positive. The first round uses the initial deficits one, and subsequent rounds include the positive finite-step deficits δ_1, ε_1.

Taking the auxiliary updates against the vectors in (48) yields

$$\begin{aligned} d_x \cdot (\widehat{\alpha}_i - \widehat{\alpha}_{i-1}) &= -\eta w \cdot (y_i - q) = \eta^2 t(|\beta_i|)(D_{i-1} + \delta_{i-1}), \\ d_y \cdot (\widehat{\beta}_i - \widehat{\beta}_{i-1}) &= \eta v \cdot (x_i - p) = \eta^2 t(|\alpha_i|)(H_{i-1} + \varepsilon_{i-1}). \end{aligned} \tag{52}$$

Each sum diverges harmonically by (15). Therefore both auxiliary dual norms tend to infinity. Bounded forward entropies to the pure pair (p, q) imply $\widehat{x}_k \to p$ and $\widehat{y}_k \to q$, and (18) transfers these limits to the played states.

The range arguments do not assume that A is invertible. In the present zero-slack singleton configuration, $A = 0$ is impossible; rank one is also incompatible with (47). These facts follow from the hypotheses, rather than being additional restrictions.

The rank configurations now exhaust all games. Restoring matrices through (8) proves Theorem 1. If $A = 0$ in another configuration, each block follows its fixed linear gradient: a nonzero one selects its pure best response, and a zero one keeps its maximally mixed state. In particular, a zero payoff leaves both sequences constant. No conclusion about a rate or a general entropy-divergence limit was used to obtain the state limits.

6 A positive limit of forward relative entropy

The preceding state-convergence theorem does not imply that forward relative entropy tends to zero. We give an exact orbit exhibiting this distinction. The observable is the one used for the Euclidean OGDA lower bound in Su et al. [5, Section 4.4, Theorem 4.9]; here we apply the raw entropy update (5) to that same game.

Put

$$P = \frac{I_2 + \sigma_z}{2}, \qquad U_0 = \frac{(I_2 - \sigma_z) \otimes \sigma_x - \sigma_x \otimes (I_2 - \sigma_z)}{4}. \tag{53}$$

Proposition 6.1. For the observable (53), the unique Nash pair on the full complex density domains is (P, P), its value is zero, and both equilibrium slack matrices vanish. Take any fixed $0 < \eta < 1/10$ and the maximally mixed initialization in (5). Both entire state sequences converge to (P, P), all finite states are positive definite, and the first two played states of either player do not commute.

There is a common limit h_η for all four one-player forward entropies:

$$\begin{gathered} D(P\|X_k),\ D(P\|Y_k),\ D(P\|\widehat{X}_k),\ D(P\|\widehat{Y}_k) \longrightarrow h_\eta, \\ \frac{\tau^2}{\log 2 + 3\tau + 2\tau \tanh \tau} \le h_\eta \le \log 2, \qquad \tau = \frac{\eta}{4}. \end{gathered} \tag{54}$$

Thus the summed forward entropy of either two-player sequence tends to $2h_\eta > 0$, although its state norm error tends to zero.

Proof. For full Bloch coordinates (x, y, z) and (s, t, w), direct evaluation gives

$$\begin{aligned} f(\rho, \sigma) &= \frac{(1 - z)s - x(1 - w)}{4}, \\ \nabla_\rho f &= \frac{s(I_2 - \sigma_z) - (1 - w)\sigma_x}{4}, \\ -\nabla_\sigma f &= \frac{x(I_2 - \sigma_z) - (1 - z)\sigma_x}{4}. \end{aligned} \tag{55}$$

The spectrum of U_0 is $\{-1/\sqrt{2}, 0, 0, 1/\sqrt{2}\}$. For a trace-zero difference with Bloch coordinates (dx, dy, dz), a corresponding full gradient-block difference has squared Frobenius

norm $dx^2/4 + dz^2/8$, at most half the squared Frobenius norm of the input. Hence $L = 1$ is valid for the full signed gradient, including its identity components.

Maximizing and minimizing over the full Bloch balls gives

$$\max_{\sigma} f(\rho, \sigma) = \frac{\sqrt{x^2 + (1-z)^2} - x}{4} \geq 0, \quad \min_{\rho} f(\rho, \sigma) = \frac{s - \sqrt{s^2 + (1-w)^2}}{4} \leq 0. \quad (56)$$

Equality forces respectively $\rho = P$ and $\sigma = P$. Both gradients in (55) vanish at this pair. This proves uniqueness, the zero value and the zero slacks, so Theorem 1 already gives the asserted state convergence.

We now track the forward entropy without invoking boundary continuity. For a two-dimensional score $h = (a, b)$, write

$$\begin{aligned} Q(h) &= \frac{\exp(a\sigma_x + b\sigma_z)}{\operatorname{Tr}\exp(a\sigma_x + b\sigma_z)}, \qquad T(h) = \frac{\tanh|h|}{|h|} h, \\ H(h) &= D(P\|Q(h)) = \log(2\cosh|h|) - b \\ &= |h| - b + \log(1 + e^{-2|h|}). \end{aligned} \quad (57)$$

Use $T(0) = 0$. Identity components in (55) cancel in each trace normalizer. Induction therefore shows that both players agree, their σ_y coordinate remains zero, and their actual played and auxiliary scores satisfy

$$\begin{aligned} h_{k+1} &= q_k + \tau v(h_k), \qquad q_{k+1} = q_k + \tau v(h_{k+1}), \\ h_0 &= q_0 = 0, \qquad v(h) = (1 - T_2(h), T_1(h)). \end{aligned} \quad (58)$$

All score coordinates are nonnegative. On this quadrant each coordinate of v lies in $[0, 1]$, and its first is strictly positive at every finite score. All finite matrices $Q(h)$ are positive definite.

In particular the exact first updates are

$$h_1 = (\tau, 0), \qquad q_1 = (\tau, \tau\tanh\tau), \qquad h_2 = (2\tau, 2\tau\tanh\tau). \quad (59)$$

With $c_2 = \tanh|h_2|/|h_2| > 0$ and $[\sigma_x, \sigma_z] = -2i\sigma_y$, one obtains

$$[Q(h_1), Q(h_2)] = -i\tau c_2 \tanh^2\tau\, \sigma_y \neq 0. \quad (60)$$

Thus the invariant real plane is not a common diagonal basis.

Lemma 1, applied to the Nash pair (P, P), gives

$$2H(q_k) \leq E_k((P,P)) \leq E_0((P,P)) = 2\log 2, \qquad H(q_k) \longrightarrow h_\eta \in [0, \log 2]. \quad (61)$$

Here symmetry identifies the comparator divergence with $2H(q_k)$, and the remaining energy term is nonnegative. Equation (21) gives $|h_k - q_k| \to 0$. Since $|\nabla H| \leq 2$, the played entropies have the same limit. Notice that (61) bounds each finite auxiliary entropy; it does not assert $H(h_k) \leq \log 2$ at every played index.

It remains to exclude $h_\eta = 0$. Write $q_k = (A_k, B_k)$, $R_k = |q_k|$ and $C = \log 2$. Equation (57) and (61) imply

$$R_k - B_k \le C, \qquad D_k := 2B_k + C + 3\tau > 0. \tag{62}$$

For the predictor $h_{k+1} = (a, b)$, its two increments from q_k lie in $[0, \tau]$. Thus $a \ge A_k$, $b \le B_k + \tau$, and the triangle inequality gives

$$|h_{k+1}| - b \le R_k - B_k + (a - A_k) \le C + \tau, \qquad |h_{k+1}| + b \le D_k. \tag{63}$$

The predictor has $a > 0$, including at $k = 0$. Its radius $R = |h_{k+1}|$ is therefore positive, and the actual corrector increments obey

$$\frac{\Delta A_k}{\Delta B_k} = \frac{R \coth R - b}{a} \ge \frac{R - b}{a} = \frac{a}{R + b} \ge \frac{A_k}{D_k}, \qquad \Delta B_k > 0. \tag{64}$$

For $k \ge 1$, put $\xi_k = \Delta B_k / D_k \ge 0$. Since $D_{k+1} = D_k + 2\Delta B_k$, the exact score ratio is nondecreasing:

$$\frac{A_{k+1}^2}{D_{k+1}} \ge \frac{A_k^2 (1 + \xi_k)^2}{D_k (1 + 2\xi_k)} \ge \frac{A_k^2}{D_k}. \tag{65}$$

Finally, $R_k + B_k \le 2B_k + C < D_k$, so for every $k \ge 1$,

$$H(q_k) \ge R_k - B_k = \frac{A_k^2}{R_k + B_k} \ge \frac{A_k^2}{D_k} \ge \frac{A_1^2}{D_1} = \frac{\tau^2}{C + 3\tau + 2\tau \tanh \tau} > 0. \tag{66}$$

Passing to the common limit proves (54). No division by A_0 or by the zero initial radius is used. ∎

This conclusion concerns $D(P\|Q)$, directed from the equilibrium to the current state. Its displayed lower bound is not a formula for h_η; it depends on the fixed step and vanishes as that step tends to zero. In particular it does not assert failure of state convergence or failure to select the unique equilibrium. The limit cannot be passed through the singular matrix logarithm: $D(P\|P) = 0$ while the forward entropies along this orbit tend to $h_\eta > 0$.

Boundary nonreciprocity was already identified in [4, Remark 4]; positive-limit full-rank paths are explicit in [9, Example 11]. Related quantum entropy conservation and recurrence were proved by Jain, Piliouras and Sim [7, Theorems 4.2–4.3] for continuous-time quantum replicator dynamics with a full-rank Nash pair. Their discrete Theorem 3.5 concerns a normalized one-step entropy change as the step tends to zero. Proposition 6.1 instead treats the fixed-step raw optimistic orbit above, with a singular zero-slack equilibrium. It uses the existing observable (53), rather than claiming a new game or a new general conservation principle.

6.1 Exact state and gap rates on the same orbit

The positive ordinary entropy limit determines the rate of the actual states. For $z = (X, Y)$ define the full-domain Nash gap by

$$\mathcal{G}(z) = \max_{V \in \mathcal{D}_2} f(X, V) - \min_{W \in \mathcal{D}_2} f(W, Y). \tag{67}$$

Proposition 6.2. Under the exact game, step and initialization of Proposition 6.1, put

$$C_\eta = \left(\frac{16 h_\eta}{3\eta}\right)^{1/3} > 0. \tag{68}$$

The complete played and auxiliary sequences satisfy

$$k^{1/3} \|z_k - (P, P)\|_{F,\mathrm{prod}} \longrightarrow C_\eta, \qquad k^{1/3} \|\widehat{z}_k - (P, P)\|_{F,\mathrm{prod}} \longrightarrow C_\eta, \tag{69}$$

$$k\mathcal{G}(z_k) \longrightarrow \frac{h_\eta}{3\eta}, \qquad k\mathcal{G}(\widehat{z}_k) \longrightarrow \frac{h_\eta}{3\eta}. \tag{70}$$

These limits hold for each fixed $0 < \eta < 1/10$. No rate for the entropy's convergence to h_η is assumed.

Proof. Retain the actual scores $q_k = (A_k, B_k)$ and $h_k = (a_k, b_k)$ from (58), and put $R_k = |q_k|$. Since $T(q_k) \to (0, 1)$, one has $\tanh R_k \to 1$, hence $R_k \to \infty$ and $B_k/R_k \to 1$. The ordinary entropy limit and (57) then give

$$R_k - B_k \longrightarrow h_\eta, \qquad \frac{A_k^2}{B_k} = \frac{(R_k - B_k)(R_k + B_k)}{B_k} \longrightarrow 2h_\eta. \tag{71}$$

In particular $A_k, B_k \to \infty$, with positive signs fixed by the invariant quadrant.

Played-state convergence gives $v(h_k) \to 0$. The literal predictor satisfies $h_{k+1} - q_k = \tau v(h_k) \to 0$, so its two coordinates are relatively asymptotic to A_k, B_k. The actual corrector increment is therefore

$$\Delta B_k := B_{k+1} - B_k = \tau T_1(h_{k+1}) \sim \tau \sqrt{2h_\eta}\, B_k^{-1/2}. \tag{72}$$

Thus $\Delta B_k / B_k \to 0$. For $k \geq 1$, $B_k > 0$ and $\Delta B_k > 0$. Setting $u_k = \sqrt{B_{k+1}/B_k} > 1$ gives the exact discrete identity

$$\frac{B_{k+1}^{3/2} - B_k^{3/2}}{\sqrt{B_k}\, \Delta B_k} = \frac{u_k^2 + u_k + 1}{u_k + 1} \longrightarrow \frac{3}{2}. \tag{73}$$

Equations (72)–(73) imply $B_{k+1}^{3/2} - B_k^{3/2} \to c := (3\tau/2)\sqrt{2h_\eta}$. Summing these increments and taking a Cesàro mean gives $B_k^{3/2}/k \to c$. With $c_B = c^{2/3}$ and $c_A = \sqrt{2h_\eta c_B}$, it follows that

$$B_k \sim b_k \sim c_B k^{2/3}, \qquad A_k \sim a_k \sim c_A k^{1/3}. \tag{74}$$

The auxiliary A_k limit follows from (71); the played limits follow from $h_k = q_{k-1} + \tau v(h_{k-1})$ and $(k-1)/k \to 1$. This keeps both actual indices unchanged.

For either $s_k = q_k$ or $s_k = h_k$, put $r_k = |s_k|$, $T(s_k) = (x_k, \zeta_k)$ and $\delta_k = 1 - \zeta_k$. Equation (74) gives $r_k/k^{2/3} \to c_B$ and

$$k^{1/3} x_k \longrightarrow \frac{c_A}{c_B} = C_\eta, \qquad k^{2/3}\delta_k \longrightarrow \frac{h_\eta}{c_B} = \frac{C_\eta^2}{2}. \tag{75}$$

To verify the second limit without assuming an entropy rate, use $H(s_k) \to h_\eta$ in (57) to obtain $r_k - s_{k,2} \to h_\eta$. The exact decomposition

$$\delta_k = \frac{r_k - s_{k,2}}{r_k} + (1 - \tanh r_k)\frac{s_{k,2}}{r_k} \tag{76}$$

then suffices: $1 - \tanh r_k \le 2e^{-2r_k}$ and $r_k \sim c_B k^{2/3}$ make the second term negligible after multiplication by $k^{2/3}$. The identities $c_A^2 = 2h_\eta c_B$ and $c_B^{3/2} = (3\tau/2)\sqrt{2h_\eta}$ give all constants in (75).

For each such equal-player pair, the product Frobenius distance is exactly $\sqrt{x_k^2 + \delta_k^2}$. Since $\delta_k/x_k \to 0$, this proves (69). The full-complex-domain best responses (56) give

$$\mathcal{G}((Q(s_k), Q(s_k))) = \frac{\sqrt{x_k^2 + \delta_k^2} - x_k}{2} = \frac{\delta_k^2}{2(\sqrt{x_k^2 + \delta_k^2} + x_k)} \sim \frac{x_k^3}{16}. \tag{77}$$

Indeed $\delta_k/x_k^2 \to 1/2$ by (75). Therefore the gap constant is $C_\eta^3/16 = h_\eta/(3\eta)$, proving (70) for both sequences. ∎

The norm in (69) is the two-player product norm; one player's Frobenius error has the additional factor $1/\sqrt{2}$. The gap is not the payoff error: the symmetric orbit has $f(Q(s_k), Q(s_k)) = 0$ at every index. These are fixed-step asymptotics, not bounds uniform over games or step sizes, and they do not determine a rate for $H(s_k) - h_\eta$.

For the same observable and maximally mixed start, the Euclidean OGDA result in [5, Theorem 4.9] gives powers k^{-1} and k^{-3} for the same product norm and Nash gap. The matrix stepsize there is also η, but Frobenius projection produces a Bloch increment with coefficient $\eta/2$, whereas (58) has entropy-score coefficient $\eta/4$. The source's separate Section 5 OMMWU rate theorem concerns a different, strictly complementary game. The present comparison retains each algorithm's own update and normalization.

7 Strict complementarity in arbitrary dimensions

This section allows arbitrary finite positive dimensions d, m and any Hermitian observable U on $\mathbb{C}^d \otimes \mathbb{C}^m$. No bound on $\|U\|_{\mathrm{op}}$ is required. Write $\mathcal{D}_n = \{Q \succeq 0 : \operatorname{Tr} Q = 1\}$ for complex Hermitian density matrices, and now set

$$\mathcal{Z} = \mathcal{D}_d \times \mathcal{D}_m, \qquad f(X, Y) = \operatorname{Tr}(U(X \otimes Y)), \qquad F = (\nabla_X f, -\nabla_Y f). \tag{78}$$

Gradients and norms are the full ambient Frobenius ones, with the product norm for pairs. Choose a Lipschitz upper bound $L \ge 1$ for F on $\mathcal{Z}$ and a fixed $0 < \eta < 1/(10L)$.

With the normalized exponential (4), used separately in each block, the exact iteration is

$$z_{k+1} = P_\eta(\widehat{z}_k, F(z_k)), \qquad \widehat{z}_{k+1} = P_\eta(\widehat{z}_k, F(z_{k+1})), \qquad z_0 = \widehat{z}_0 = (I_d/d, I_m/m). \tag{79}$$

Both stages have the same old auxiliary base. The first stage uses the already available $F(z_k)$; only $F(z_{k+1})$ is newly evaluated.

Theorem 2. *Suppose there exists one Nash pair $z_* = (X_*, Y_*)$ of value v whose slacks satisfy*

$$\begin{aligned} S_X &= \nabla_X f(z_*) - vI_d \succeq 0, \quad \ker S_X = \operatorname{range} X_*, \\ S_Y &= vI_m - \nabla_Y f(z_*) \succeq 0, \quad \ker S_Y = \operatorname{range} Y_*. \end{aligned} \tag{80}$$

Then both complete sequences in (79) *converge in product Frobenius norm to a common Nash pair $z_\infty = (X_\infty, Y_\infty)$. Every finite iterate is positive definite. Moreover,*

$$\begin{aligned} \operatorname{range} X_\infty = \operatorname{range} X_*, \qquad \operatorname{range} Y_\infty = \operatorname{range} Y_*, \\ D(z_\infty \| z_k) \longrightarrow 0, \qquad D(z_\infty \| \widehat{z}_k) \longrightarrow 0, \end{aligned} \tag{81}$$

where D is the sum of the two quantum relative entropies.

Only existence of one strictly complementary pair is assumed. It is not an input to the algorithm, and the selected pair need not equal that witness or itself be strictly complementary. The theorem requires neither uniqueness nor full rank, and gives no prescribed selection formula, convergence rate or finite-length conclusion. Its entropy comparator is the selected limit. Strict complementarity is a sufficient hypothesis for this arbitrary-dimensional result; it is not necessary for point convergence, as Theorem 1 and Proposition 6.1 demonstrate.

For densities $R \succeq 0$, $Q \succ 0$, define

$$D(R\|Q) = \operatorname{Tr}(R \log R) - \operatorname{Tr}(R \log Q), \qquad 0 \log 0 = 0. \tag{82}$$

The first term is interpreted on the support of R. The logarithms below are only taken at positive definite finite iterates. We prove the theorem directly, including the boundary steps needed for (81).

7.1 An energy for every comparator

Bilinearity and the signed gradient convention give the exact identity

$$\langle F(z) - F(u), z - u \rangle = 0, \qquad g_u(z) := \langle F(z), z - u \rangle = \langle F(u), z - u \rangle. \tag{83}$$

For $R = P_\eta(Q, G)$, the logarithmic update is $\log R = \log Q - \eta G - cI$, where $c = \log \operatorname{Tr} \exp(\log Q - \eta G)$. Expanding the divergences gives, for every density W, including singular ones,

$$D(W\|R) = D(W\|Q) - D(R\|Q) - \eta \langle G, R - W \rangle. \tag{84}$$

We shall use

$$D(R\|Q) \geq \tfrac{1}{2}\|R-Q\|_F^2. \tag{85}$$

Indeed, in an eigenbasis of a positive density, the Hessian of $\operatorname{Tr} Q\log Q$ has diagonal coefficients $1/\lambda_i$ and off-diagonal divided differences $(\log\lambda_i-\log\lambda_j)/(\lambda_i-\lambda_j)$. They are all at least one because $0<\lambda_i\leq 1$. Integration along a segment proves (85) for positive R; continuity in R gives the boundary case. Addition gives the pair version.

Put

$$t=\eta L<\tfrac{1}{10},\quad A_k=D(z_{k+1}\|\widehat{z}_k),\quad B_k=D(\widehat{z}_k\|z_k),\quad \mathcal{E}_k(u)=D(u\|\widehat{z}_k)+tB_k. \tag{86}$$

Here $B_0=0$. Applying (84) to the two updates with their common base gives

$$\begin{aligned} D(u\|\widehat{z}_{k+1}) = D(u\|\widehat{z}_k) &- \eta g_u(z_{k+1}) - A_k - B_{k+1} \\ &+ \eta\langle F(z_{k+1})-F(z_k), z_{k+1}-\widehat{z}_{k+1}\rangle. \end{aligned} \tag{87}$$

If $a=\|z_{k+1}-\widehat{z}_k\|$, $b=\|\widehat{z}_k-z_k\|$ and $c=\|z_{k+1}-\widehat{z}_{k+1}\|$, the last term is at most $t(a+b)c$. Young's inequality and (85) bound it by $tA_k+tB_k+2tB_{k+1}$. Thus, for *every* comparator $u\in\mathcal{Z}$,

$$\mathcal{E}_{k+1}(u)\leq \mathcal{E}_k(u)-\eta g_u(z_{k+1})-(1-t)A_k-(1-3t)B_{k+1}. \tag{88}$$

This is an estimate for the actual indexing in (79).

Let P_X, P_Y be the support projections of the witness, and use $i\in\{X,Y\}$ for a block. Define

$$m(z)=\sum_i \operatorname{Tr}(P_i^{\perp} z_i). \tag{89}$$

For the witness, $g_{z_*}(z)=\operatorname{Tr}(S_X X)+\operatorname{Tr}(S_Y Y)\geq 0$. Each nonempty orthogonal complement has a positive slack eigenvalue bound $S_i\succeq \delta_i P_i^{\perp}$. If a complement is empty, its leakage is zero and no δ_i is needed. The nonnegative decreasing energy in (88) therefore yields

$$\sum_{k\geq 0} A_k<\infty,\qquad \sum_{k\geq 1} B_k<\infty,\qquad \sum_{k\geq 1} m(z_k)<\infty. \tag{90}$$

In particular $z_k,\widehat{z}_k,z_{k+1}$ have pairwise distances tending to zero.

Auxiliary leakage is also summable. For any projection T and densities R,Q, with $Q\succ 0$, one has

$$\operatorname{Tr}(TR)\leq D(R\|Q)+(e-1)\operatorname{Tr}(TQ). \tag{91}$$

To see this, nonnegativity of the divergence from R to the normalized exponential of $\log Q+T$ gives $D(R\|Q)\geq \operatorname{Tr}(TR)-\log\operatorname{Tr} e^{\log Q+T}$. Golden–Thompson and $e^T=I+(e-1)T$ bound that logarithm by $\log(1+(e-1)\operatorname{Tr}(TQ))\leq (e-1)\operatorname{Tr}(TQ)$. Apply (91) blockwise with $R=\widehat{z}_{i,k}$ and $Q=z_{i,k}$. Equations (90) give

$$\sum_{k\geq 1} m(\widehat{z}_k)<\infty. \tag{92}$$

All cluster states have exactly the witness supports. The leakage first puts their support inside P_i. To prove the reverse inclusion, write $h(z_*) = \sum_i \operatorname{Tr}(z_{*,i} \log z_{*,i})$ and $C = \mathcal{E}_0(z_*) - h(z_*)$. Bounded comparator entropy gives $\sum_i \operatorname{Tr}(z_{*,i}(-\log \widehat{z}_{i,k})) \le C$. Since $-\log \widehat{z}_{i,k} \succeq 0$ and $z_{*,i} \succeq \mu_i P_i$ for some $\mu_i > 0$, scalar Jensen on a unit vector in range P_i gives

$$P_i \widehat{z}_{i,k} P_i \succeq e^{-C/\mu_i} P_i. \tag{93}$$

This passes to every cluster, and the vanishing distances give the same conclusion for the played states. It establishes support and interiority on the face, not yet Nash optimality.

Finally let $M = \max_{i,z\in\mathcal{Z}} \|F_i(z)\|_{\mathrm{op}} < \infty$. The literal initialized recursion gives

$$\widehat{z}_{i,k} = \frac{e^{H_{i,k}}}{\operatorname{Tr} e^{H_{i,k}}}, \qquad H_{i,k} = -\eta \sum_{j=1}^{k} F_i(z_j), \tag{94}$$

and $z_{i,k+1}$ has score $H_{i,k} - \eta F_i(z_k)$. Consequently, with $n_X = d$, $n_Y = m$,

$$\|\log \widehat{z}_{i,k}\|_{\mathrm{op}} \le 2\eta M k + \log n_i, \qquad \|\log z_{i,k+1}\|_{\mathrm{op}} \le 2\eta M(k+1) + \log n_i. \tag{95}$$

No commutation or smallness of ηM is used.

7.2 Two boundary lemmas

The first lemma detects gradient blocks crossing a limiting support. Write $[A, B] = AB - BA$.

Lemma 3. *For densities $R \succeq 0$ and $Q \succ 0$,*

$$D(R\|Q) \ge \frac{\|[R, \log Q]\|_F^2}{8(1 + \|\log Q\|_{\mathrm{op}})}. \tag{96}$$

Proof. Put $A = \log Q$, $C_0 = [R, A]$ and $K = \mathrm{i}C_0$. The matrix K is Hermitian. For $R_s = e^{\mathrm{i}sK} R e^{-\mathrm{i}sK}$ and $b(s) = \operatorname{Tr}(R_s A)$, cyclicity gives

$$b'(0) = -\operatorname{Tr}(C_0^2) = \|C_0\|_F^2, \qquad b''(s) = -\operatorname{Tr}([K, [K, R_s]]A). \tag{97}$$

Expanding the double commutator and using $0 \preceq R_s \preceq I$ gives $|b''(s)| \le 4\|A\|_{\mathrm{op}}\|K\|_F^2$. For example $\|K^2 R_s\|_1 \le \|K^2\|_1 \|R_s\|_{\mathrm{op}} \le \|K\|_F^2$, and the term $K R_s K \succeq 0$ has trace at most $\|K\|_F^2$; the remaining term is identical. Taking $s = [4(1 + \|A\|_{\mathrm{op}})]^{-1}$ in Taylor's integral bound yields $b(s) - b(0) \ge \|C_0\|_F^2/[8(1 + \|A\|_{\mathrm{op}})]$. Unitary conjugation preserves $\operatorname{Tr} R \log R$, while $D(R_s\|Q) \ge 0$. Hence $D(R\|Q) \ge D(R\|Q) - D(R_s\|Q) = b(s) - b(0)$, proving the claim. ∎

Lemma 4. *Suppose positive definite densities Q_j converge to q, with support projection P, and $k_j \to \infty$ satisfies*

$$\|\log Q_j\|_{\mathrm{op}} = O(k_j), \qquad k_j \operatorname{Tr}(P^{\perp} Q_j) \longrightarrow 0. \tag{98}$$

Then, as operators on range P*,*

$$P \log Q_j P \longrightarrow \log_P q, \qquad D(q\|Q_j) \longrightarrow 0. \tag{99}$$

Proof. Let $r = \operatorname{rank} P$ and let Π_j project onto the largest r eigenvalues of Q_j. These eigenvalues are eventually at least some $\alpha > 0$, and the others tend to zero. Since $Q_j \succeq \alpha \Pi_j$ and P, Π_j have the same rank,

$$\operatorname{Tr}(P(I - \Pi_j)) = \operatorname{Tr}(P^\perp \Pi_j) \le \alpha^{-1} \operatorname{Tr}(P^\perp Q_j). \tag{100}$$

The small-eigenvalue contribution to $P(-\log Q_j)P$ is positive semidefinite. Its norm is at most its trace, which is bounded by $\|\log Q_j\|_{\mathrm{op}} \operatorname{Tr}(P(I - \Pi_j)) \to 0$. For the large eigenvalues choose a continuous scalar function that equals $\log x$ on their eventual interval and zero near zero. Continuous functional calculus then gives convergence of that part to $\log_P q$, extended by zero on $P^\perp$. This proves the compressed-log limit; taking its trace against q proves the entropy limit. When $P = I$, the small-eigenvalue part is absent and ordinary logarithm continuity suffices. ∎

7.3 A selected cluster and the complete trajectory

We complete the proof of Theorem 2. All finite states are positive definite because every update is a normalized exponential of a finite Hermitian matrix. The estimates above are therefore valid at every index.

The nonnegative sequence

$$a_k = A_k + B_k + B_{k+1} + m(\widehat{z}_k) + m(z_{k+1}) + m(z_k) \quad (k \ge 1) \tag{101}$$

is summable. It has a subsequence $k_j \to \infty$ with $k_j a_{k_j} \to 0$: otherwise it would eventually dominate a positive multiple of the harmonic sequence. By compactness and the vanishing residuals, pass to a further subsequence such that

$$\widehat{z}_{k_j} \longrightarrow q, \qquad z_{k_j} \longrightarrow q, \qquad z_{k_j+1} \longrightarrow q. \tag{102}$$

Each block q_i has exactly support P_i by (93). Lemma 4, applied to both $\widehat{z}_{i,k_j}$ and z_{i,k_j+1} with (95), gives convergence of their compressed logarithms to the same $\log_{P_i} q_i$, and in particular

$$D(q\|\widehat{z}_{k_j}) \longrightarrow 0. \tag{103}$$

Compress the exact first-stage logarithmic identity:

$$P_i(\log z_{i,k+1} - \log \widehat{z}_{i,k})P_i = -\eta P_i F_i(z_k) P_i - c_{i,k} P_i. \tag{104}$$

Its left side tends to zero along k_j. Taking the normalized trace on P_i identifies the limit of c_{i,k_j}; substituting back proves

$$P_i F_i(q) P_i = b_i P_i \tag{105}$$

for a scalar b_i. This alone would not control the off-support blocks. Since $z_{i,k+1}$ commutes with its own logarithm, the same exact update gives

$$[z_{i,k+1}, \log \widehat{z}_{i,k}] = \eta [z_{i,k+1}, F_i(z_k)]. \tag{106}$$

Lemma 3 and (95) therefore imply, for a fixed finite C_1,

$$\sum_i \|[z_{i,k+1}, F_i(z_k)]\|_F^2 \le C_1 (k+1) A_k. \tag{107}$$

Along k_j this tends to zero. Hence $[q_i, F_i(q)] = 0$. In its block from P_i to $P_i^\perp$, this identity is $(q_i|_{P_i}) P_i F_i(q) P_i^\perp = 0$. Invertibility on P_i and Hermiticity give

$$P_i F_i(q) P_i^\perp = 0, \qquad P_i^\perp F_i(q) P_i = 0. \tag{108}$$

Thus $T_i := F_i(q) - b_i I$ is supported on $P_i^\perp$. Using (83) and unit trace, for every $z \in \mathcal{Z}$,

$$g_q(z) = \sum_i \mathrm{Tr}(T_i z_i), \qquad |g_q(z)| \le C_q m(z), \qquad C_q = \max_i \|T_i\|_{\mathrm{op}}. \tag{109}$$

No sign of T_i is assumed here, and q has not yet been proved Nash.

Use the universal energy (88) with comparator q. Its possible positive error is bounded by the summable sequence $\eta C_q m(z_{k+1})$. More explicitly,

$$\mathcal{E}_k(q) + \eta C_q \sum_{j=k+1}^{\infty} m(z_j) \tag{110}$$

is nonnegative and nonincreasing. Along k_j it tends to zero by (103), $B_{k_j} \to 0$ and tail summability. Thus $D(q\|\widehat{z}_k) \to 0$ along the entire sequence. Equation (85) and $B_k \to 0$ imply

$$\widehat{z}_k \longrightarrow q, \qquad z_k \longrightarrow q. \tag{111}$$

The played entropy also tends to zero: the exact identity (84) gives

$$D(q\|z_{k+1}) = D(q\|\widehat{z}_k) - A_k - \eta \langle F(z_k), z_{k+1} - q \rangle \longrightarrow 0, \tag{112}$$

using boundedness of F and (111). No continuity assertion at a general singular boundary is needed.

It remains to establish full best-response inequalities, including exterior slack signs. Fix *any* $u \in \mathcal{Z}$. Its initial entropy is finite. Summing (88) and dropping nonnegative energies and dissipations gives

$$\eta \sum_{k=0}^{N-1} \langle F(z_{k+1}), z_{k+1} - u \rangle \le \mathcal{E}_0(u). \tag{113}$$

Only now, after (111), divide by ηN and pass to the limit. We obtain

$$\langle F(q), u - q \rangle \ge 0 \qquad (u \in \mathcal{Z}). \tag{114}$$

Taking arbitrary deviations in each block proves full-domain Nash optimality. All claims of Theorem 2 follow with $z_\infty = q$. The argument includes full-support and one-dimensional blocks, whose complementary leakage is identically zero. It does not assert that reverse divergences $D(z_k\|z_\infty)$ are finite when the limit is singular.

The distinction from the qubit argument is substantive. Under (80), summable mass outside the witness supports and an actual-iterate commutator estimate restore entropy reciprocity along a suitable subsequence, then the signed comparator energy selects the entire trajectory. Without strict complementarity those summability conclusions need not follow, and the fixed orbit of Section 6 has a positive forward entropy limit. The theorem does not extend the unconditional qubit conclusion to arbitrary dimensions without (80).

8 Selection on a non-strict qutrit face

The preceding arbitrary-dimensional theorem does not apply when every Nash pair fails strict complementarity. The following fixed game shows selected-point convergence on a nonunique singular face outside that hypothesis. Alice now has a three-level state and Bob a qubit; their full complex domains are $\mathcal{D}_3 \times \mathcal{D}_2$.

Let E_{ij} denote three-by-three matrix units and put $S_{ij} = E_{ij} + E_{ji}$. Define

$$\begin{gathered} B_0 = E_{33} + S_{13}, \quad B_1 = S_{13} + S_{23}, \quad B_2 = \mathrm{i}(E_{23} - E_{32}), \quad B_3 = -S_{13}, \\ U_1 = \tfrac{1}{8}\big(B_0 \otimes I_2 + B_1 \otimes \sigma_x + B_2 \otimes \sigma_y + B_3 \otimes \sigma_z\big). \end{gathered} \tag{115}$$

Use $f(X,Y) = \mathrm{Tr}(U_1(X \otimes Y))$, with Alice minimizing, Bob maximizing, and the exact raw recurrence (79) from $(I_3/3, I_2/2)$. Set

$$P_A = \mathrm{diag}(1,1,0), \qquad P_B = \mathrm{diag}(1,0), \qquad p = (P_A/2, P_B). \tag{116}$$

Proposition 8.1. The observable satisfies $\|U_1\|_F^2 = 11/32 < 1$, and $L = 1$ is a valid Lipschitz bound for its full signed gradient. Its full-domain Nash set, of value zero, is exactly

$$\{X \in \mathcal{D}_3 : \mathrm{range}\, X \subseteq \mathrm{range}\, P_A\} \times \{P_B\}. \tag{117}$$

There is no strictly complementary Nash pair. Nevertheless, for every fixed $0 < \eta < 1/10$, both complete canonically initialized sequences satisfy

$$z_k \longrightarrow p, \qquad \widehat{z}_k \longrightarrow p, \qquad D(p\|z_k) \longrightarrow 0, \qquad D(p\|\widehat{z}_k) \longrightarrow 0. \tag{118}$$

Every finite state is positive definite. The actual first and third played Alice states have only scalar matrices in their common commutant. Moreover, the only local unitary pairs $V \in U(3), W \in U(2)$ satisfying $f(VXV^*, WYW^*) = f(X,Y)$ for every pair of density matrices are scalar unitaries.

The asserted selection concerns this one fixed observable and initialization. It is neither a general non-strict-complementarity theorem nor a universal maximum-entropy selection rule. The commutant conclusion excludes a fixed qubit block with a passive spectator for this actual trajectory; it makes no assertion about arbitrary nonlinear changes of variables.

8.1 Full-domain geometry and the actual score form

For $Y = (I_2 + x\sigma_x + y\sigma_y + \zeta\sigma_z)/2$, direct contraction gives

$$8\nabla_X f(Y) = \begin{pmatrix} 0 & 0 & 1-\zeta+x \\ 0 & 0 & x+\mathrm{i}y \\ 1-\zeta+x & x-\mathrm{i}y & 1 \end{pmatrix}. \tag{119}$$

The squared entries of $8U_1$ sum to 22. In orthonormal Hermitian bases, the payoff's coefficient matrix represents $Y \mapsto \nabla_X f$, and its transpose represents $X \mapsto \nabla_Y f$. Its operator norm is at most its Frobenius norm $\|U_1\|_F < 1$, proving the stated Lipschitz bound.

Every Alice state supported on P_A has zero payoff against every Bob state, while $f(X, P_B) = X_{33}/8 \geq 0$. This proves value zero. Optimality of X forces $X_{33} = 0$, and positivity then forces $X_{13} = X_{23} = 0$. Conversely, Bob optimality requires (119) to be positive semidefinite. A positive semidefinite matrix with a zero diagonal entry has the corresponding row and column zero. Thus $x + \mathrm{i}y = 0$ and $1 - \zeta + x = 0$, giving $Y = P_B$. This proves (117) over the full complex domains. The slacks are $E_{33}/8$ for Alice and zero for Bob; Bob's rank-one state and zero slack cannot satisfy strict complementarity.

Finite-time positivity follows from normalized exponentials. Real symmetric states form an invariant set from the given start. For their Bob Bloch coordinates (x, ζ) and real Alice entries, define

$$c(Y) = (1 - \zeta + x, x), \qquad b(X) = (2(X_{13} + X_{23}), -2X_{13}), \qquad \tau = \eta/8. \tag{120}$$

The real trajectory need not commute, and is not used to restrict best responses. Scalar identity components of Bob's gradient cancel; its traceless part is $b(X) \cdot (\sigma_x, \sigma_z)/8$.

Use auxiliary scores $v_k, q_k \in \mathbb{R}^2$ and played scores $v'_k, q'_k \in \mathbb{R}^2$. With all initial scores zero, their exact recurrences for $k \geq 1$ are

$$\begin{aligned} v'_k &= v_{k-1} - \tau c(Y_{k-1}), & q'_k &= q_{k-1} + \tau b(X_{k-1}), \\ v_k &= v_{k-1} - \tau c(Y_k), & q_k &= q_{k-1} + \tau b(X_k). \end{aligned} \tag{121}$$

The actual auxiliary states are

$$\widehat{X}_k = \frac{e^{H_k}}{\operatorname{Tr} e^{H_k}}, \quad H_k = \begin{pmatrix} 0 & 0 & v_{k,1} \\ 0 & 0 & v_{k,2} \\ v_{k,1} & v_{k,2} & -\tau k \end{pmatrix}, \qquad \widehat{Y}_k = \frac{e^{q_{k,1}\sigma_x + q_{k,2}\sigma_z}}{2\cosh|q_k|}. \tag{122}$$

Replacing v_k, q_k by their primed versions gives the played states, with the same exterior entry $-\tau k$ at index k. These formulas are exact summations of the raw updates.

8.2 Entropy selection at the singular face

We use the universal energy (88) and mass-transfer inequality (91); their derivations do not require strict complementarity. Here $t = \eta$, $\mathcal{E}_0(p) = \log 3$, and

$$\langle F(z), z - p\rangle = f(X, P_B) - f(P_A/2, Y) = X_{33}/8. \tag{123}$$

Thus $\mathcal{E}_k(p)$ decreases, $\sum_k A_k < \infty$, $\sum_{k\geq 1} B_k < \infty$, and

$$\sum_{k\geq 1} m_k < \infty, \qquad \sum_{k\geq 1} \widehat{m}_k < \infty, \qquad m_k = (X_k)_{33}, \quad \widehat{m}_k = (\widehat{X}_k)_{33}. \tag{124}$$

The auxiliary sum follows by applying (91) to E_{33}. Also $D(p\|\widehat{z}_k) = \mathcal{E}_k(p) - \eta B_k$ has an ordinary finite limit. We will prove that its Cesàro mean is zero, without assuming entropy continuity at a singular limit.

The eigenvalues of H_k are $0, \lambda_k, -\tau k - \lambda_k$, where

$$\lambda_k = \frac{\sqrt{\tau^2 k^2 + 4|v_k|^2} - \tau k}{2} \geq 0, \qquad |v_k|^2 = \lambda_k(\tau k + \lambda_k). \tag{125}$$

Writing $Z_k = 1 + e^{\lambda_k} + e^{-\tau k - \lambda_k}$ gives the exact singular-comparator identity

$$L_k := D(P_A/2\|\widehat{X}_k) = \log(Z_k/2). \tag{126}$$

Indeed the compression of $\log \widehat{X}_k$ onto P_A is exactly $-\log Z_k I_2$. Nonnegativity of the two block divergences gives $L_k \leq \log 3$, hence $Z_k \leq 6$ and $0 \leq \lambda_k \leq K := \log 6$. In particular $|v_k| = O(\sqrt{k})$.

For $k \geq 1$, the exterior population is exactly

$$\widehat{m}_k = \frac{\lambda_k e^{\lambda_k} + (\tau k + \lambda_k) e^{-\tau k - \lambda_k}}{(\tau k + 2\lambda_k) Z_k}. \tag{127}$$

This formula includes $\lambda_k = 0$. Since $Z_k \leq 3e^{\lambda_k}$, $\lambda_k \leq 3(\tau k + 2K)\widehat{m}_k$. Moreover $L_k \leq \lambda_k + \frac{1}{2}e^{-\tau k}$. Consequently

$$0 \leq L_k \leq C(k+1)\widehat{m}_k + \tfrac{1}{2}e^{-\tau k}, \qquad \frac{1}{N}\sum_{k=1}^{N} L_k \longrightarrow 0. \tag{128}$$

For the last implication, split a finite prefix and bound the remaining normalized sum of $(k+1)\widehat{m}_k$ by twice a tail of the convergent series in (124).

Next put $\beta_k = (Y_k)_{22}$ and $\widehat{\beta}_k = (\widehat{Y}_k)_{22}$. Subtracting the two components of the actual v_k update gives

$$2\tau \sum_{j=1}^{k} \beta_j = v_{k,2} - v_{k,1} \leq \sqrt{2}|v_k| = O(\sqrt{k}). \tag{129}$$

Mass transfer with Bob's projection $I_2 - P_B$ yields $\widehat{\beta}_k \leq B_k + (e-1)\beta_k$. Hence

$$\sum_{k=1}^{N} \widehat{\beta}_k = O(\sqrt{N}). \tag{130}$$

All constants here may depend on the one fixed $\eta > 0$.

Positivity of the played Alice matrix gives $|X_{13}|^2+|X_{23}|^2 \le X_{33}$. Thus (121) and (124) imply

$$\sum_{k\ge1} |q_k - q_{k-1}|^2 < \infty, \qquad |q_k| = o(\sqrt{k}), \qquad M_N := \max_{1\le k\le N}(1+|q_k|) = o(\sqrt{N}). \tag{131}$$

For the middle statement, split the score sum at a fixed index and apply Cauchy–Schwarz to its tail; the tail square sum can be made arbitrarily small. The maximum statement follows by treating a finite prefix separately.

Apply Lemma 2, specifically (23), to Bob's pure comparator. Equivalently, the qubit entropy bound there gives a fixed C_0 such that $D(P_B\|\widehat{Y}_k) \le 2C_0(1+|q_k|)\widehat{\beta}_k$. Together with (130)–(131), this proves

$$\frac{1}{N}\sum_{k=1}^{N} D(P_B\|\widehat{Y}_k) \le \frac{2C_0 M_N}{N}\sum_{k=1}^{N}\widehat{\beta}_k \longrightarrow 0. \tag{132}$$

Equations (128) and (132) make the Cesàro mean of $D(p\|\widehat{z}_k)$ zero. Since that nonnegative quantity already has an ordinary limit, its limit is zero. Strong entropy convexity gives $\widehat{z}_k \to p$, and $B_k \to 0$ gives $z_k \to p$. Finally the exact first-stage identity gives

$$D(p\|z_{k+1}) = D(p\|\widehat{z}_k) - A_k - \eta\langle F(z_k), z_{k+1}-p\rangle \longrightarrow 0. \tag{133}$$

The last pairing tends to zero by boundedness of F and whole-state convergence. This proves all limits in (118); the averages were only a proof device.

8.3 Actual three-level activity and payoff symmetry

For completeness, the startup shows why this orbit does not reduce to a fixed qubit block. Write $\mathcal{E}(H) = e^H/\operatorname{Tr} e^H$, $H_0 = B_0$, $R = \sigma_x - \sigma_z$ and $C = 2S_{13} + S_{23}$. The first Alice off-diagonal entry is

$$r = (\mathcal{E}(-\tau H_0))_{13} = -\frac{2e^{-\tau/2}\sinh(\sqrt{5}\tau/2)}{\sqrt{5}[1+2e^{-\tau/2}\cosh(\sqrt{5}\tau/2)]} < 0. \tag{134}$$

The actual updates give

$$\begin{gathered} X_1 = \widehat{X}_1 = \mathcal{E}(-\tau H_0), \quad Y_1 = I_2/2, \quad \widehat{Y}_1 = \mathcal{E}(2\tau r R), \\ X_2 = \mathcal{E}(-2\tau H_0), \quad Y_2 = \mathcal{E}(4\tau r R), \quad u = \tfrac{1}{\sqrt{2}}\tanh(4\sqrt{2}\tau r) < 0, \\ \widehat{X}_2 = \mathcal{E}[-\tau(2H_0+uC)], \qquad X_3 = \mathcal{E}[-\tau(3H_0+2uC)]. \end{gathered} \tag{135}$$

Here Y_2 has Bloch coordinates $(u,0,-u)$, so its new Alice gradient is $(H_0+uC)/8$. It is used once in $\widehat{X}_2$ and again in X_3. In particular X_1, X_2 commute; they cannot replace the first and third states in the claim.

The spectrum of H_0 is $0, (1-\sqrt{5})/2, (1+\sqrt{5})/2$, with zero eigenvector e_2. The other eigenvectors lie in $\mathrm{span}\{e_1, e_3\}$ and have nonzero third coordinate. A matrix commuting with X_1, X_3 commutes with their logarithms, hence with H_0, C. It is diagonal in the simple-spectrum eigenbasis of H_0. Since $Ce_2 = e_3$ has nonzero components along both other eigenvectors, commutation with C forces all three diagonal entries to be equal. This proves the actual commutant assertion.

Finally an exact joint unitary payoff symmetry preserves the Nash set. Hence W fixes P_B and acts on Bob's Bloch coordinates by a rotation about the ζ axis, while $V = \mathrm{diag}(V_A, e^{\mathrm{i}\phi})$ preserves P_A. Comparing the cross column in (119), the coefficient of $1-\zeta$ forces $V_A e_1 = e^{\mathrm{i}\phi} e_1$. Unitarity makes its other basis vector an eigenvector too. The first cross component $1-\zeta+x$ then forces the Bob rotation to preserve x for every Bloch vector, so it is the identity and W is scalar. The second component $x + \mathrm{i}y$ forces the remaining Alice phase to equal $e^{\mathrm{i}\phi}$, so V is scalar. This completes Proposition 8.1. Simultaneous complex conjugation is a real-structure symmetry, but its fixed points include many states on Alice's face and do not alone select $P_A/2$.

The selected Alice state maximizes entropy on this particular Nash face. That static characterization is not used as a convergence theorem. Gunasekar et al. [10, Theorems 1 and 1a] characterize interpolating limits of mirror descent conditional on convergence in their data-fitting setting. Wu and Rebeschini [11, Theorem 2] prove a spectral-entropy selection result for ordinary unnormalized mirror descent in matrix sensing, allowing singular limits and noncommuting sensing matrices. Neither result identifies the normalized two-sequence game trajectory here. The contribution of this example is its actual canonical selection and entropy limit on a non-strict, nonunique face; no general selection principle, convergence rate or priority claim is inferred.

References

[1] F. Vasconcelos, E.-V. Vlatakis-Gkaragkounis, P. Mertikopoulos, G. Piliouras and M. I. Jordan, A quadratic speedup in finding Nash equilibria of quantum zero-sum games, Quantum 9 (2025), 1737. https://doi.org/10.22331/q-2025-05-06-1737.

[2] A. Barakat, W. Lin, J. Lazarsfeld and A. Varvitsiotis, Optimistic online learning in symmetric cone games, Transactions on Machine Learning Research (2026), author version arXiv:2504.03592v4. https://arxiv.org/abs/2504.03592v4.

[3] F. Orabona, Last-iterate convergence of optimistic multiplicative weight update, arXiv:2606.11773v1 (2026). https://arxiv.org/abs/2606.11773v1.

[4] J. López and E. A. Papa Quiroz, Construction of proximal distances over symmetric cones, Optimization 66 (2017), 1301–1321. https://doi.org/10.1080/02331934.2016.1277998.

[5] Y. Su, E.-V. Vlatakis-Gkaragkounis and P. Xiong, Average-and last-iterate lower bounds for optimistic matrix mirror-prox in quantum zero-sum games, arXiv:2609.38835v1 (2026). https://arxiv.org/abs/2609.38835v1.

[6] Y. Su, E.-V. Vlatakis-Gkaragkounis and P. Xiong, Breaking $1/\varepsilon$ barrier in quantum zero-sum games: Generalizing metric subregularity for spectraplexes, arXiv:2509.21570v2 (2026). https://arxiv.org/abs/2509.21570v2.

[7] R. Jain, G. Piliouras and R. Sim, Matrix multiplicative weights updates in quantum zero-sum games: Conservation laws and recurrence, Advances in Neural Information Processing Systems 35 (2022). https://papers.nips.cc/paper_files/paper/2022/hash/1a78459dbbcdc90783d183999e72176c-Abstract-Conference.html.

[8] S. Weis, Information topologies on non-commutative state spaces, Journal of Convex Analysis 21 (2014), 339–399; author version arXiv:1003.5671v4, Example 2.2. https://arxiv.org/abs/1003.5671v4.

[9] S. Weis and A. Knauf, Entropy distance: New quantum phenomena, Journal of Mathematical Physics 53 (2012), 102206; author version arXiv:1007.5464v3, Example 11. https://doi.org/10.1063/1.4757652.

[10] S. Gunasekar, J. Lee, D. Soudry and N. Srebro, Characterizing implicit bias in terms of optimization geometry, Proceedings of Machine Learning Research 80 (2018), 1832–1841. https://proceedings.mlr.press/v80/gunasekar18a.html.

[11] F. Wu and P. Rebeschini, Implicit regularization in matrix sensing via mirror descent, arXiv:2105.13831v2 (2021). https://arxiv.org/abs/2105.13831v2.